Satvik Bagul

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EDUCATION

The University of Massachusetts, Amherst | Amherst, MA
Double Major: Computer Science and **Statistics and Data Science | Minor: Business | GPA:** 3.3/4.0
Awards & Honors: *Dean's List* (Fall 2025), *National and Professional Champion*, ADNOC YasInSchools 4x4InSchools (2024); *Innovator's Award*, First Tech Challenge (2017); *Best delegate*, WINMUN (2024); *Honorable mention*, EISJMUN (2024)

TECHNICAL PROJECTS

Quantitative Options Research Engine | Personal Project…….Fall 2026
- Built a Python/Streamlit options analytics engine implementing *Black-Scholes-Merton* and *American CRR binomial pricing*, with continuous dividend yield, analytical Greeks, and model-price comparison.
- Developed Implied-Volatility extraction using *Brent's root-finding method*, *no-arbitrage bounds*, *convergence checks*, and numerical failure handling for invalid market prices.
- Implemented adaptive CRR step refinement and a *Jarrow-Rudd fallback* for invalid risk-neutral probabilities in short-expiry and low-volatility scenarios.
- Processed option-chain data with Bid/Ask/Mid/Spread, quote-quality and liquidity filters, separating observed market data from model outputs; vectorized pricing and Greeks calculations with NumPy.
Link: opengine.streamlit.app

Volatility Surface Intelligence Engine | Personal Project……Fall 2026
- Built a Python/Streamlit research system using **historical SPY option data** to model implied volatility across moneyness, maturity, option type, and liquidity features.
- Developed and compared **Gradient Boosting and MLP neural-network models** for out-of-sample implied-volatility prediction using chronological train/validation/test splits.
- Evaluated prediction quality through **MAE, RMSE, residual analysis, and downstream Black-Scholes pricing error**, investigating how model errors vary across the volatility surface.
Link: https://iv-mlproject.streamlit.app/

Automated Financial Scraper | Personal Project……Winter 2025
- Designed an automated financial analytics tool to extract stock data and generate actionable metrics such as moving averages (helps generate buy/sell signals), volatility, and annualized performance.
- Data extracted using *YFinance API*, GUI built using *Streamlit*, uses *pandas* to compile data, data visualization using *matplotlib*, and storage of data using *CSV files*.

Risk and Forecast Modeling System | Built for Hack(Her)-413……..Spring 2026
- Built a *full stack SaaS platform* that generates AI-powered project execution blueprints with *work breakdown structures*, *timeline forecasting*, *budget modeling*, and *risk analysis* using probability-weighted scoring.
- Implemented *Monte Carlo simulations* (1000 loops) for timeline predictions with *p10/p50/p90 confidence intervals*.
- Developed with *React*, *FastAPI*, and *MongoDB*. Integrated with *Google Gemini* for blueprint generation. Features GANTT charts, risk heatmaps, and PDF export functionality.

SimplySerenity, AI Therapist | Built for HackBrown……..Spring 2026
- Developed an AI-powered virtual therapist using *Java* with an *HTML/CSS* frontend to provide real-time, empathetic mental health support.
- Built on *Google Gemini's API* and utilizes tools from *ElevenLabs* that allow the user and AI to interact verbally.
- Integrated with *Stripe API* to provide easy payment solutions
-Link: simplyserenity.tech

ADDITIONAL EXPERIENCE

President of Student Council | Our Own High School, Al Warqa'a……April 2024- March 2025
- Provided *strategic leadership* to the student body, led council initiatives, and served as the *primary liaison* between students and school administration.

Partnerships Division and Tutor | Instilt Educate……..Dec 2024- Jan 2025
- Cultivated partnerships with *10+ educational institutions and key organizations*, including the *British Council (India)* and the *Ministry of Education (Ampara)*, also tutored students from African academic institutions.

SKILLS

- **Programming Languages:** Python, C++, Java, HTML, CSS
- **Tools and Technologies:** MS Excel, MS PowerPoint, Canva, Git